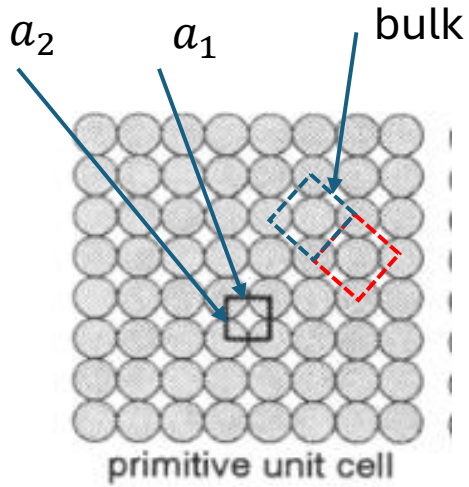
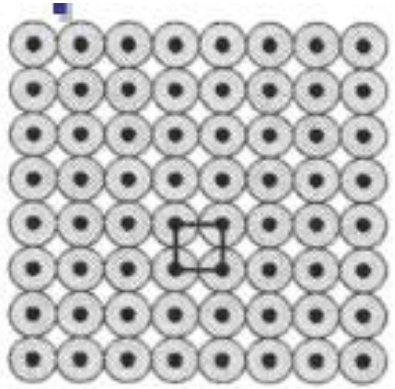
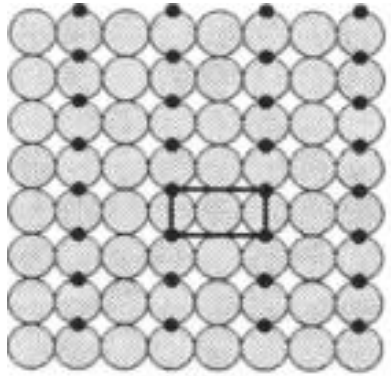


Příklady Woodove notace

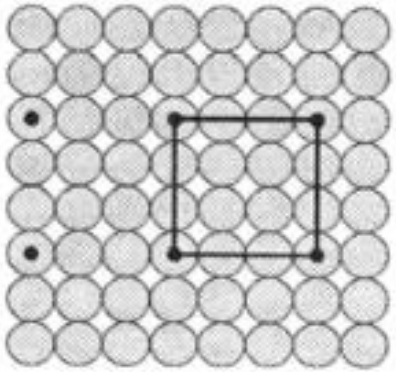
$b_1 = a_1 \quad b_2 = a_2$

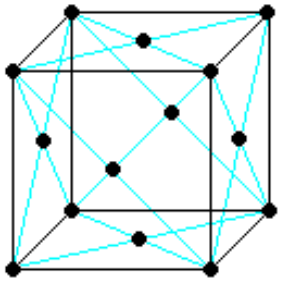


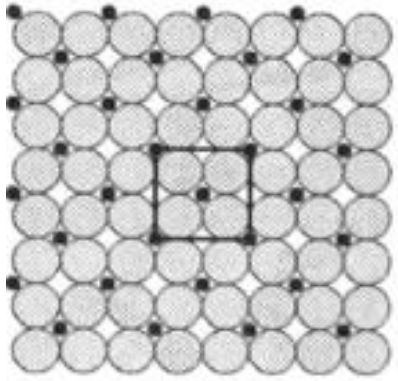
$b_1 = 2a_1 \quad b_2 = a_2$



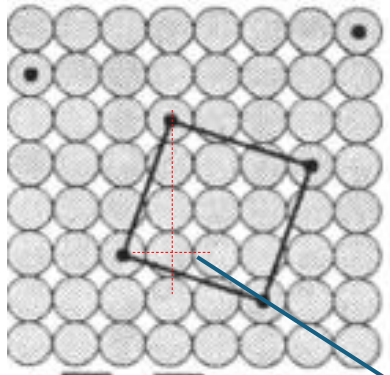
$b_1 = 3a_1 \, b_2 = 3a_2$

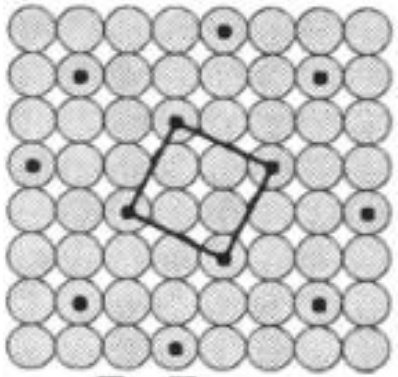


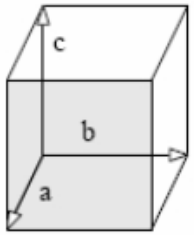




$b_1 = 2a_1 \quad b_2 = 2a_2$







fcc(100)

(h,k,l)=(1/x,1/y,1/z)

Figure 2.71 Wood's notation for a number of overlayers on the (100) face of an FCC metal.

$$b_1 = \sqrt{a_1^2 + a_2^2} = \sqrt{10}$$

$$b_2 = \sqrt{a_1^2 + a_2^2} = \sqrt{10}$$

$$\theta = \arctan\left(\frac{1}{3}\right) \approx 18.4°$$

Masel, Richard I.. "Principles of Adsorption and Reaction on Solid Surfaces." (1996).

Příklady Woodove notace

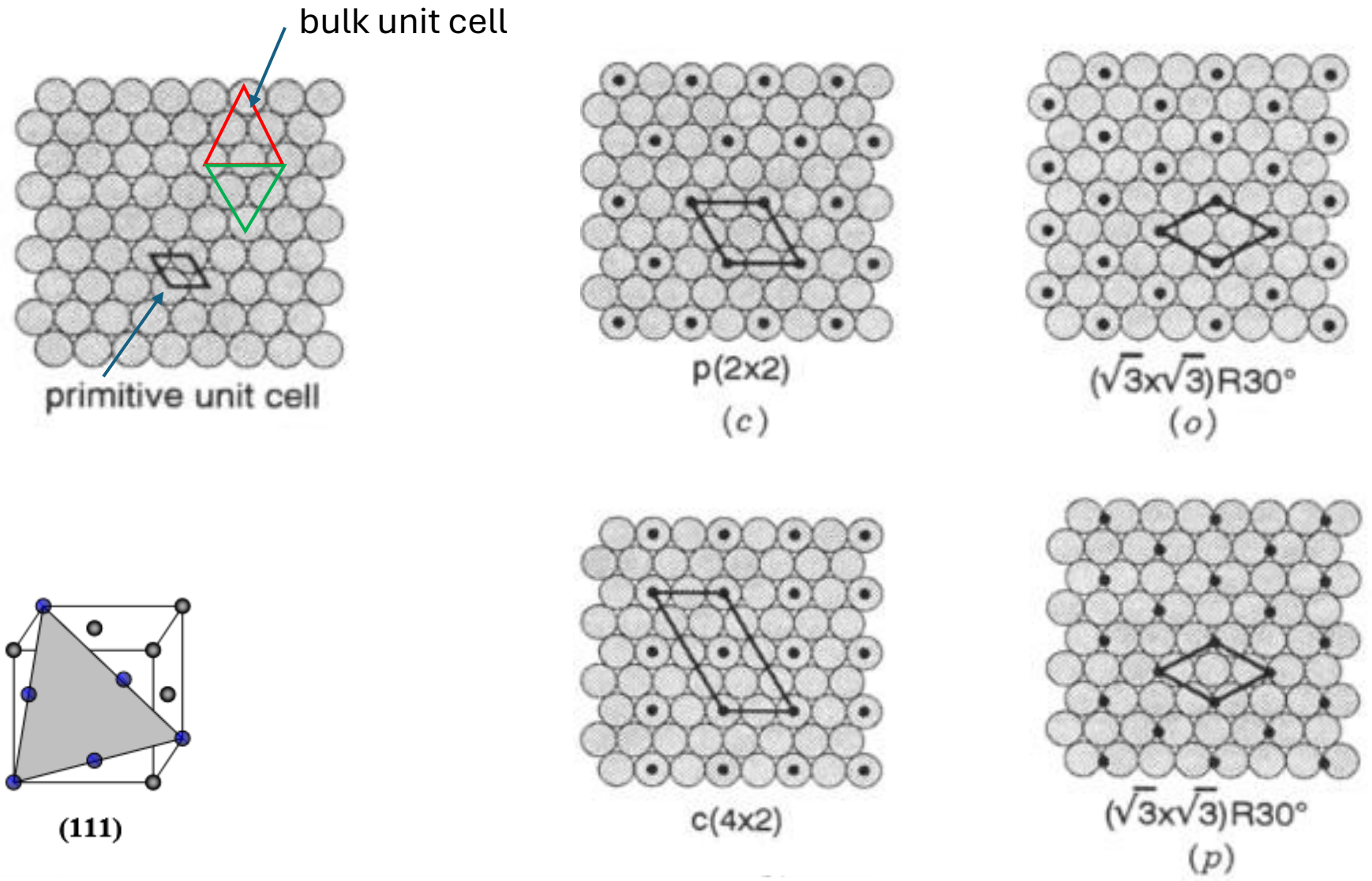


Figure 2.72 Wood's notation for a number of overlayers on the (111) face of an FCC metal.

Masel, Richard I.. "Principles of Adsorption and Reaction on Solid Surfaces." (1996).

Příklady Woodove notace

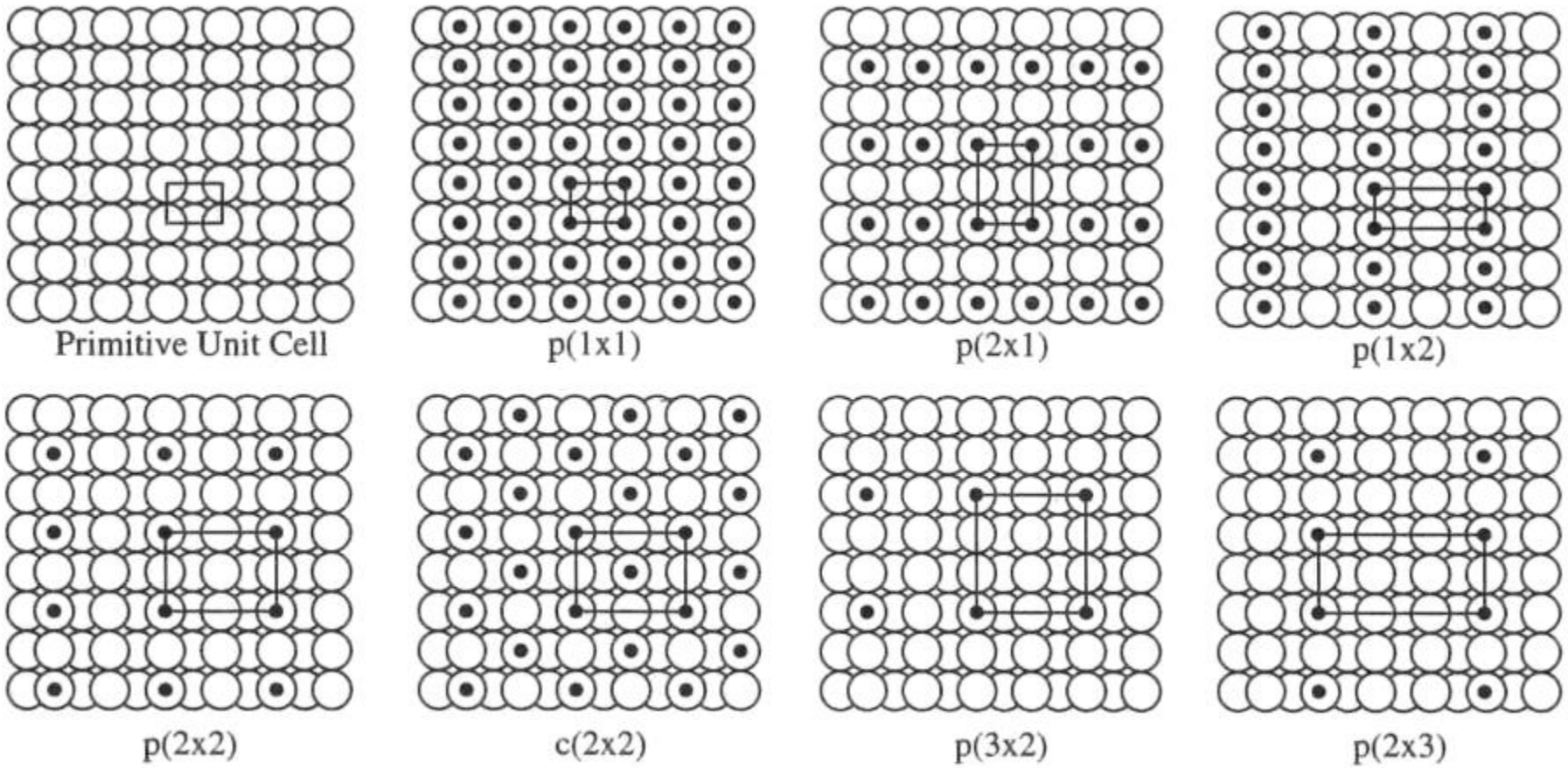


Figure 2.73 Wood's notation for a number of overlayers on the (110) face of an FCC metal.

Masel, Richard I.. "Principles of Adsorption and Reaction on Solid Surfaces." (1996).

Metody určení hustoty šroubových a hranových dislokací v polovodičových materiálech:

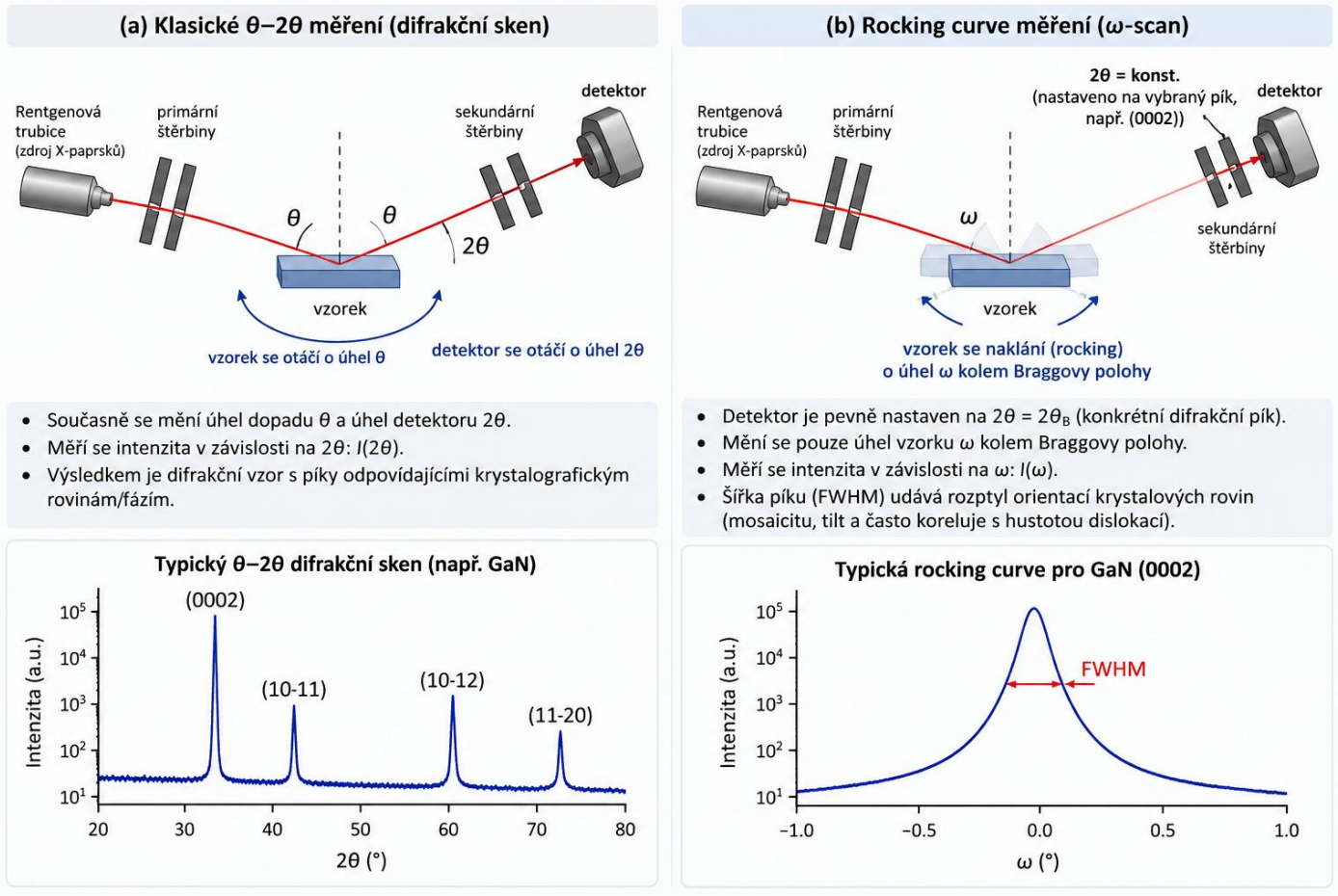
(a) **Rentgenová difrakce (XRD) –** nepřímý odhad hustoty dislokací z FWHM vybraných difrakčních reflexi;

(b) **Etch-pit density (EPD) –** selektivní leptání, při kterém se místa výstupu dislokací na povrch projeví jako leptací jamky; jejich hustota se následně určuje pomocí SEM nebo AFM jako počet jamek na jednotku plochy;

(c) **Transmisní elektronová mikroskopie (TEM) –** přímé pozorování a počítání dislokací; umožňuje také určit jejich typ;

(d) **CL/PL mapping –** některé dislokace se mohou projevit jako oblasti se sníženou luminiscenční intenzitou, často jako tmavé body, což umožňuje statistický odhad jejich hustoty.

Rentgenová difrakce (XRD)



Pro epitaxní vrstvy se často používá aproximace:

$$\rho_D = \beta^2 / (2\pi ln2 * b^2) = \beta^2 / 4.36 * b^2$$

$\boldsymbol{\beta}$-FWHM roking curve v radianech;
b-Burgersův vektor (který popisuje velikost a směr deformace krystalové mřížky způsobené dislokací; pro hranovou dislokaci $\boldsymbol{b} \perp$ dislokační čáře, pro šrobovou dislokaci je $\boldsymbol{b} \parallel$ dislokační čáře)
$\boldsymbol{4,36 \approx 2\pi ln2}$ geometricky/statisticky faktor

$$\rho_{screw} = \beta^2 / 4.35 * b_s^2$$

$$\rho_{edge} = \beta^2 / 4.35 * b_e^2$$

Princip XRD je založen na tom, že na krystalický materiál dopadá monochromatické rentgenové záření o známé vlnové délce λ. Rentgenové záření se rozptyluje na elektronech atomů v jednotlivých krystalových rovinách. Pokud se rozptýlené vlny z různých rovin sejdou ve fázi, dojde ke konstruktivní interferenci a objeví se difrakční maximum – pík.

Podmínku pro vznik píku popisuje **Braggův zákon**: $n\lambda = 2d \sin \Theta$

Etch-pit density (EPD)

GaN/safir (Kyma technogy)

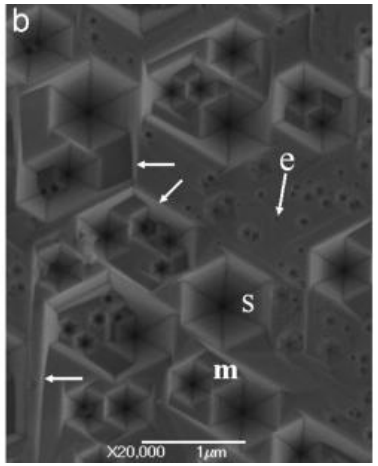
postup:

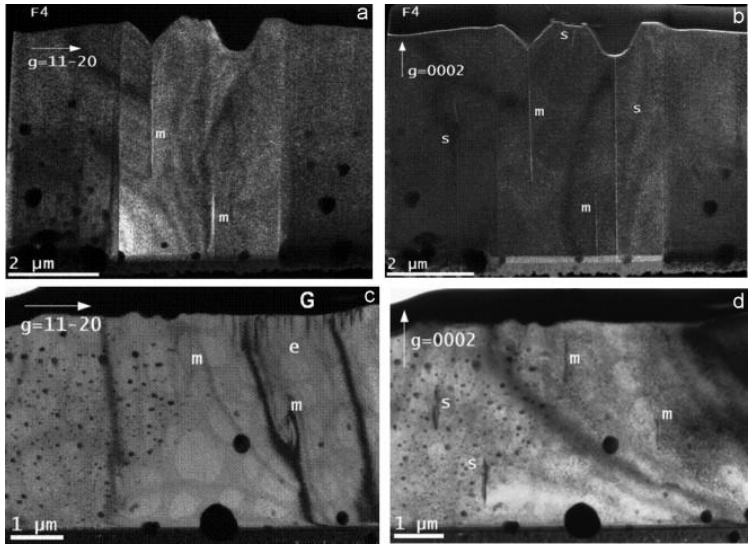
- **KOH**(hydroxid draselný)-**NaOH** (hydroxid sodny) eutectic + 10 % **MgO** (oxid hořečnatý);
- 450 °C, 5 min;
- DI+H_2O; 20-30% HCl

$$\rho_{D=}{}^{N}/_{A} \sim 4x10^{8}\ cm^{-2}$$

Transmisní elektronová mikroskopie (TEM)

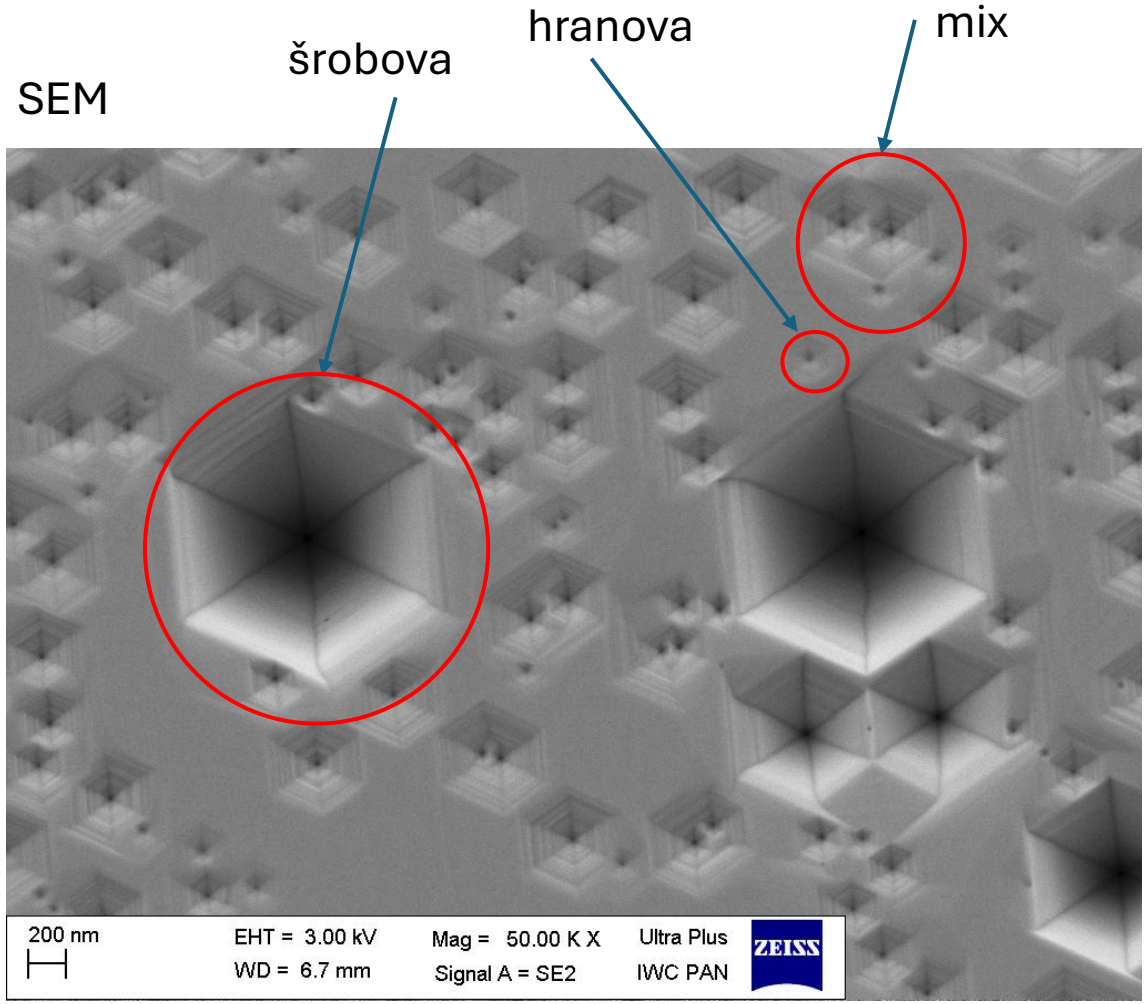
SEM





10.1016/j.jcrysgro.2007.03.030

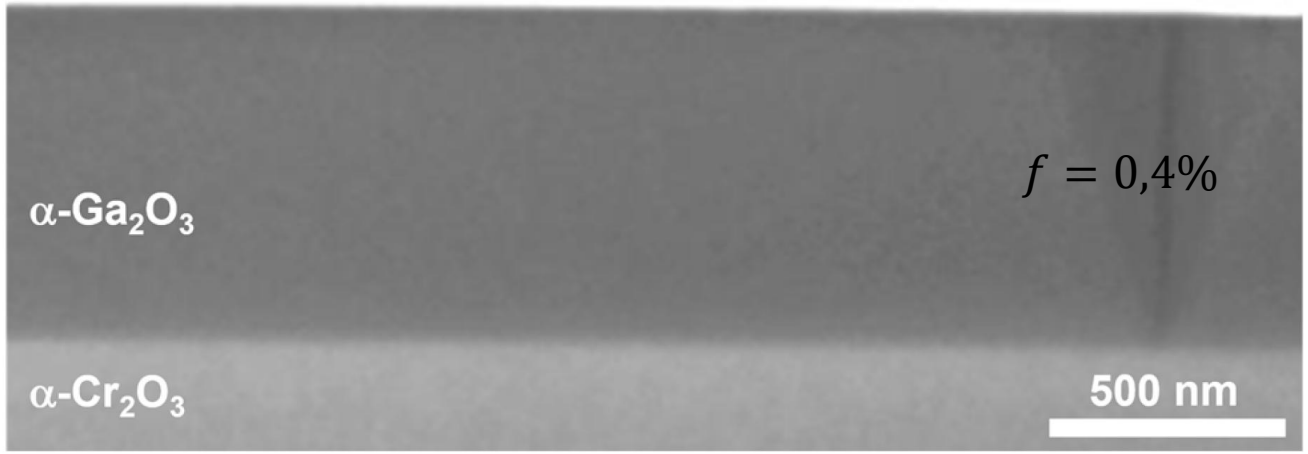
SEM



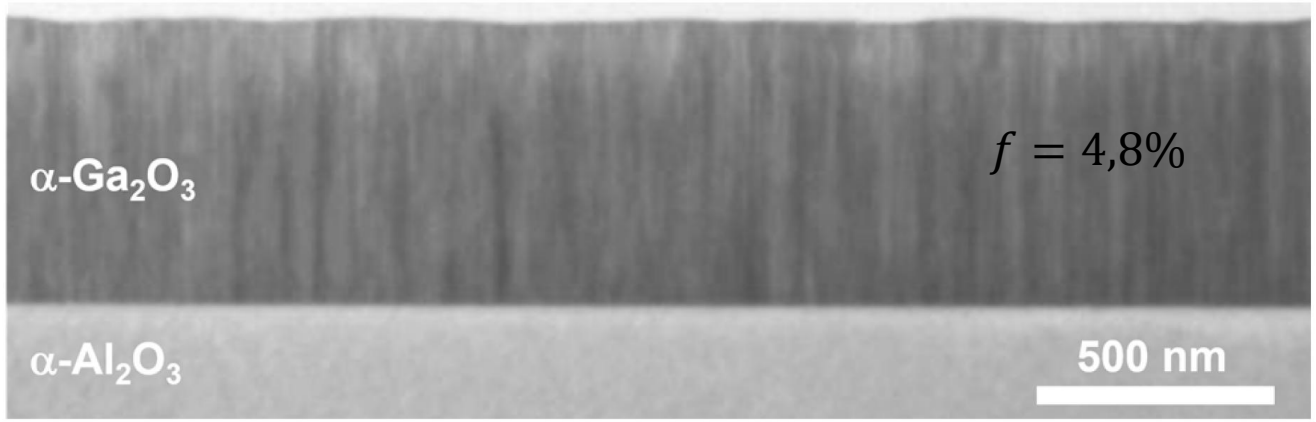
Neopublikovaná data –UFE AVCR/Nanomater

Transmisní elektronová mikroskopie (TEM)

Bright-field TEM

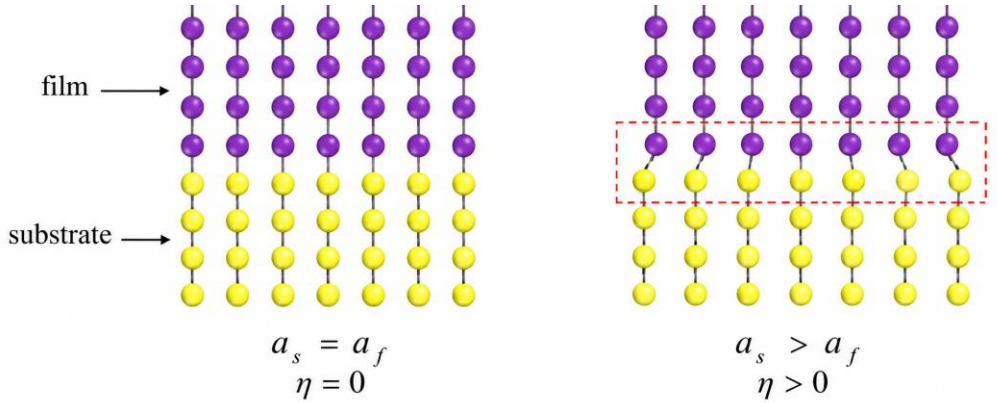


(a)



(b)

https://doi.org/10.3390/cryst16030193



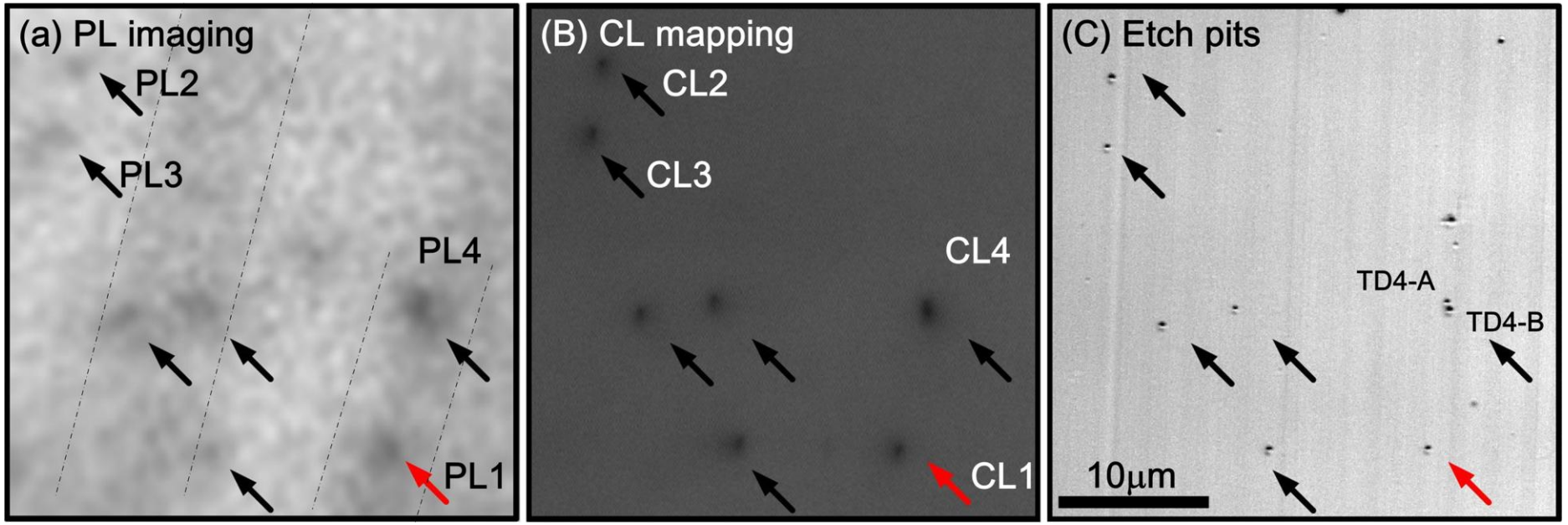
Mřížkový nesoulad (lattice misfit)

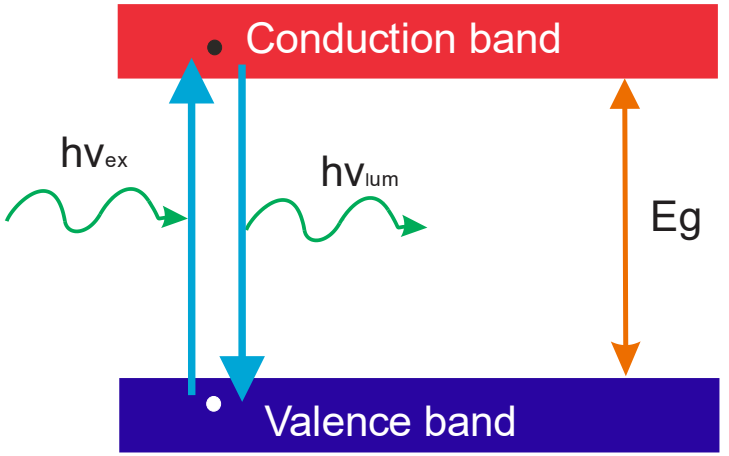
$$f = \frac{a_s - a_f}{a_f}$$

	XRC FWHM (Arcsec.)		Estimated Dislocation Density (cm^{-2})	
	0006	$10\bar{1}4$	Screw	Edge
α-Ga_2O_3 film on α-Cr_2O_3 template	105	135	3×10^6	7×10^7
	(95)	(117)	(3×10^6)	(6×10^7)
α-Ga_2O_3 film on α-Al_2O_3 substrate	51	3016	8×10^5	3×10^{10}

CL/PL mapping

https://doi.org/10.1016/j.jcrysgro.2017.01.017



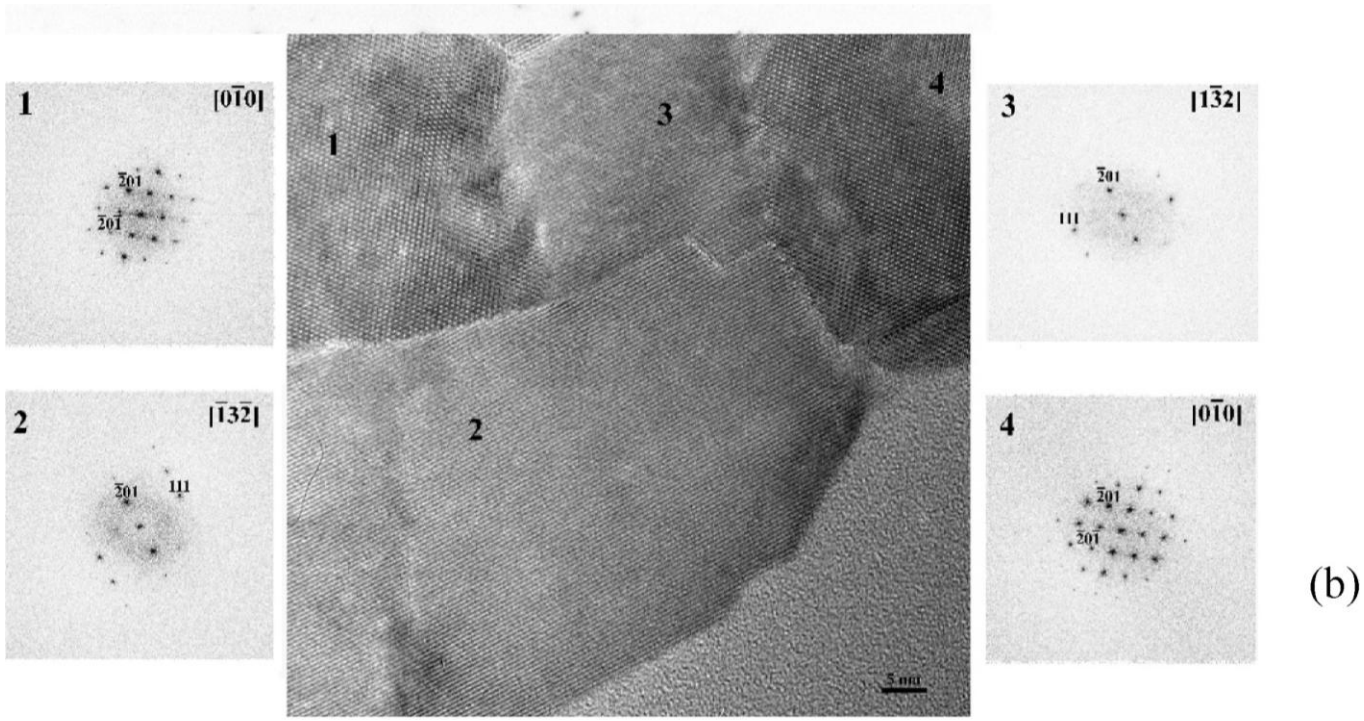


Určovani plošnych poruch v polovodičových materiálech:

Hranice zrn

TEM Transmission Electron Microscopy.

β-Ga_2O_3



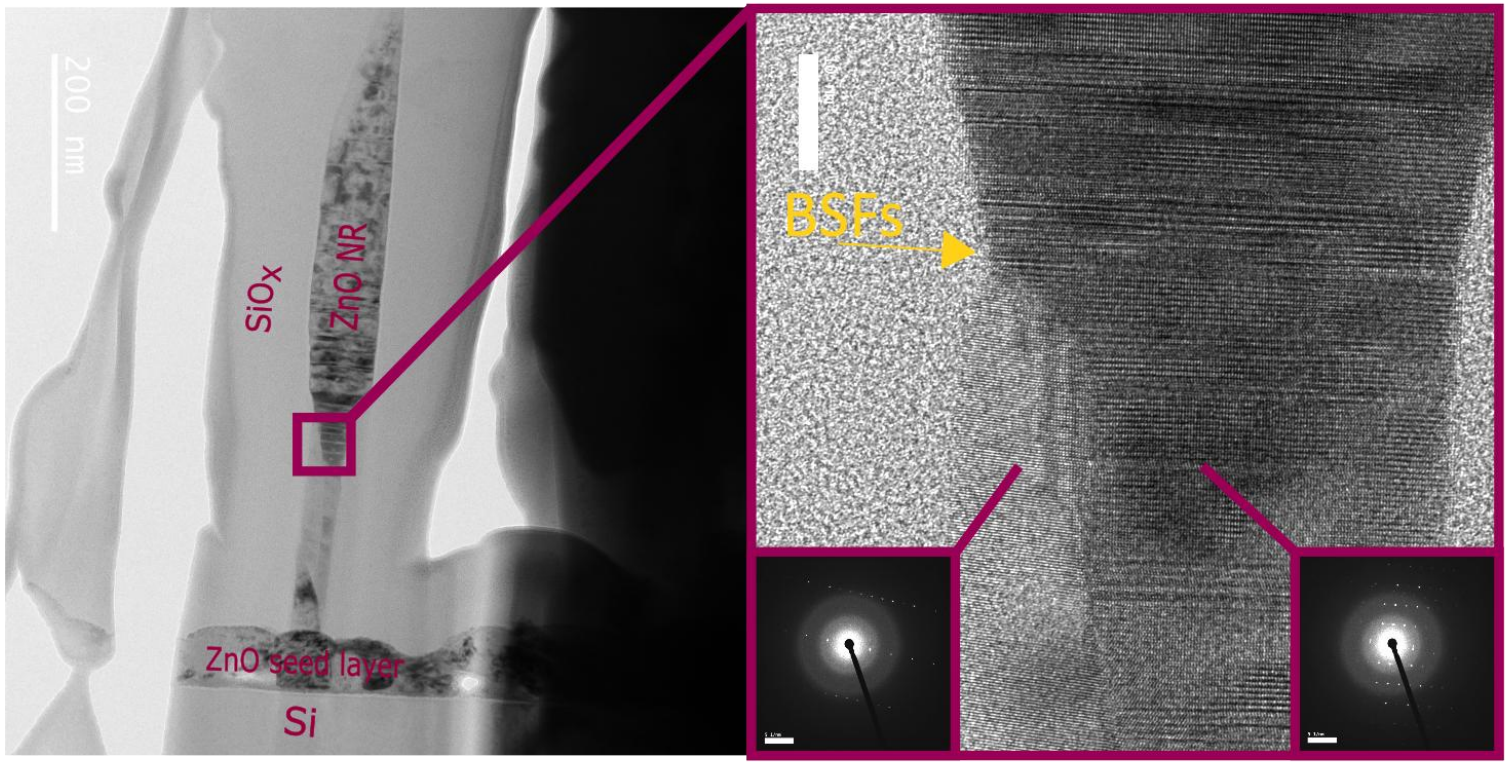
(b)

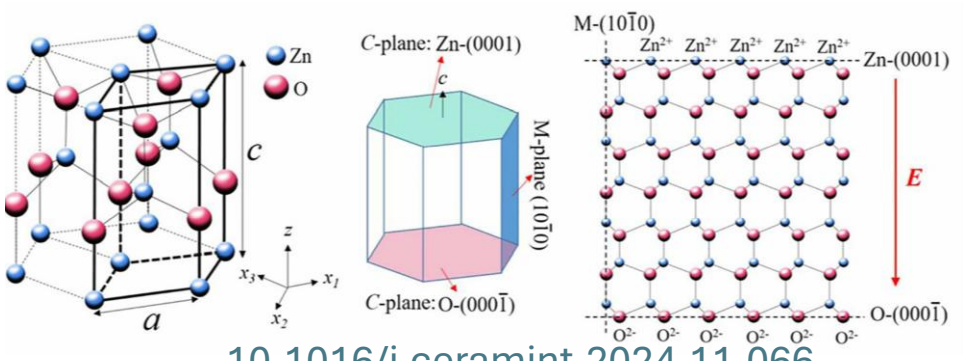
https://doi.org/10.1016/j.actamat.2017.08.062

Určovani plošné poruchy v polovodičových materiálech:

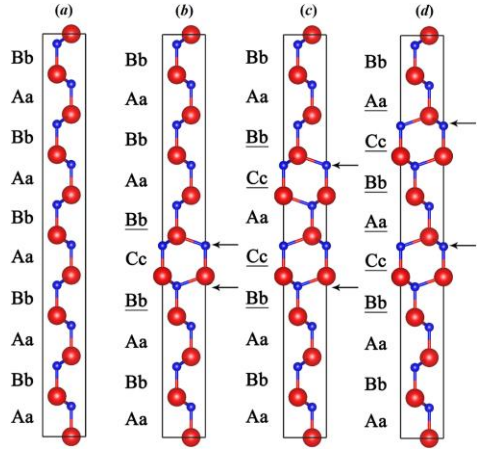
Vrstevná porucha (stacking fault) je porucha krystalové mřížky, při které je narušeno správné pořadí atomových rovin.

TEM Transmission Electron Microscopy.

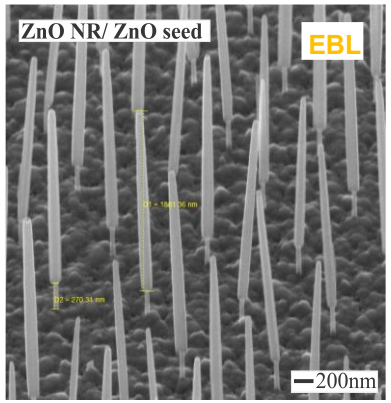


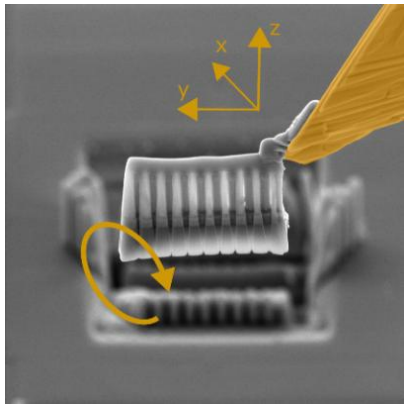


10.1016/j.ceramint.2024.11.066



https://doi.org/10.1016/j.cocom.2025.e01033

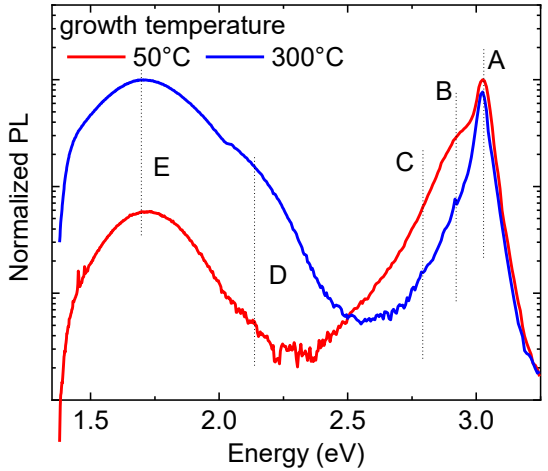


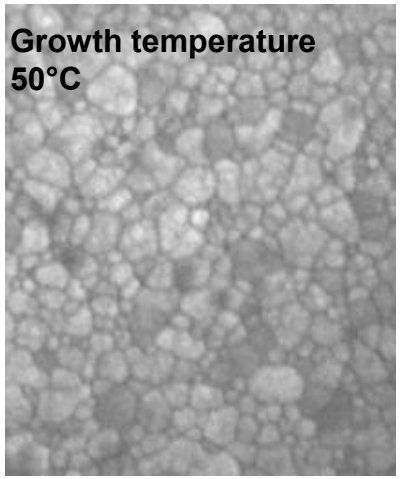


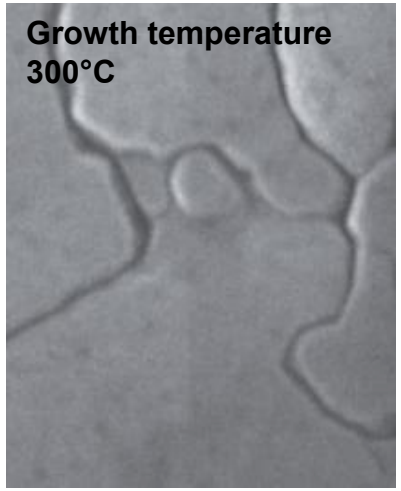
Určovani bodovych poruchy v polovodičových materiálech:

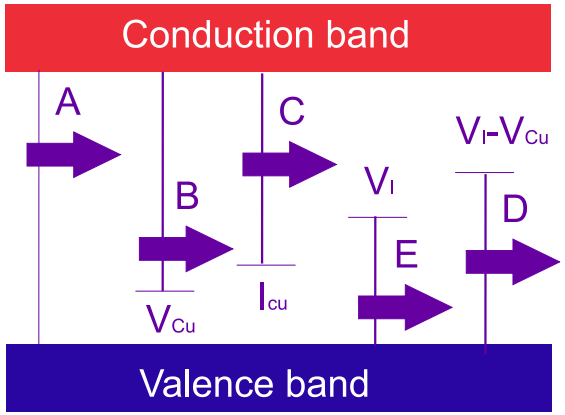
Fotoluminiscenční spektroskopie (PL)

CuI (PLD)

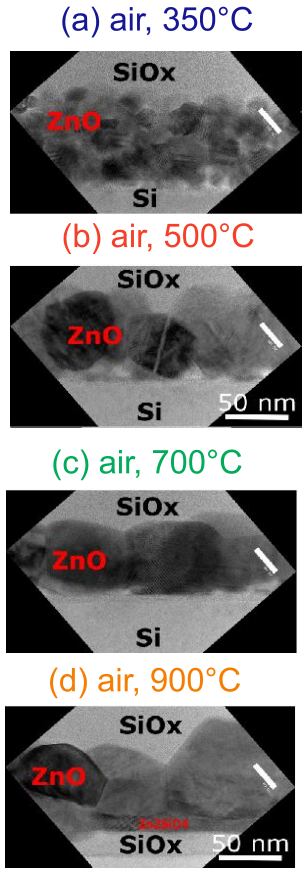


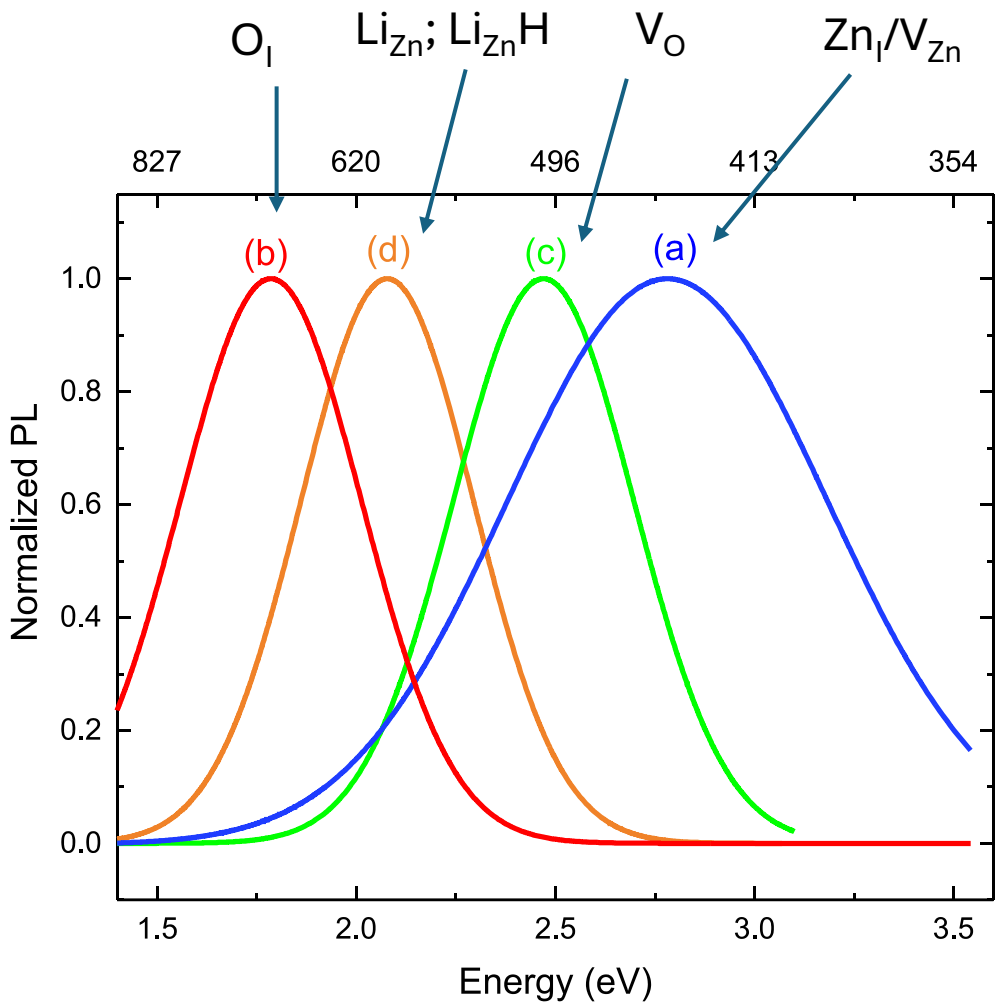






ZnO (sol-gel)



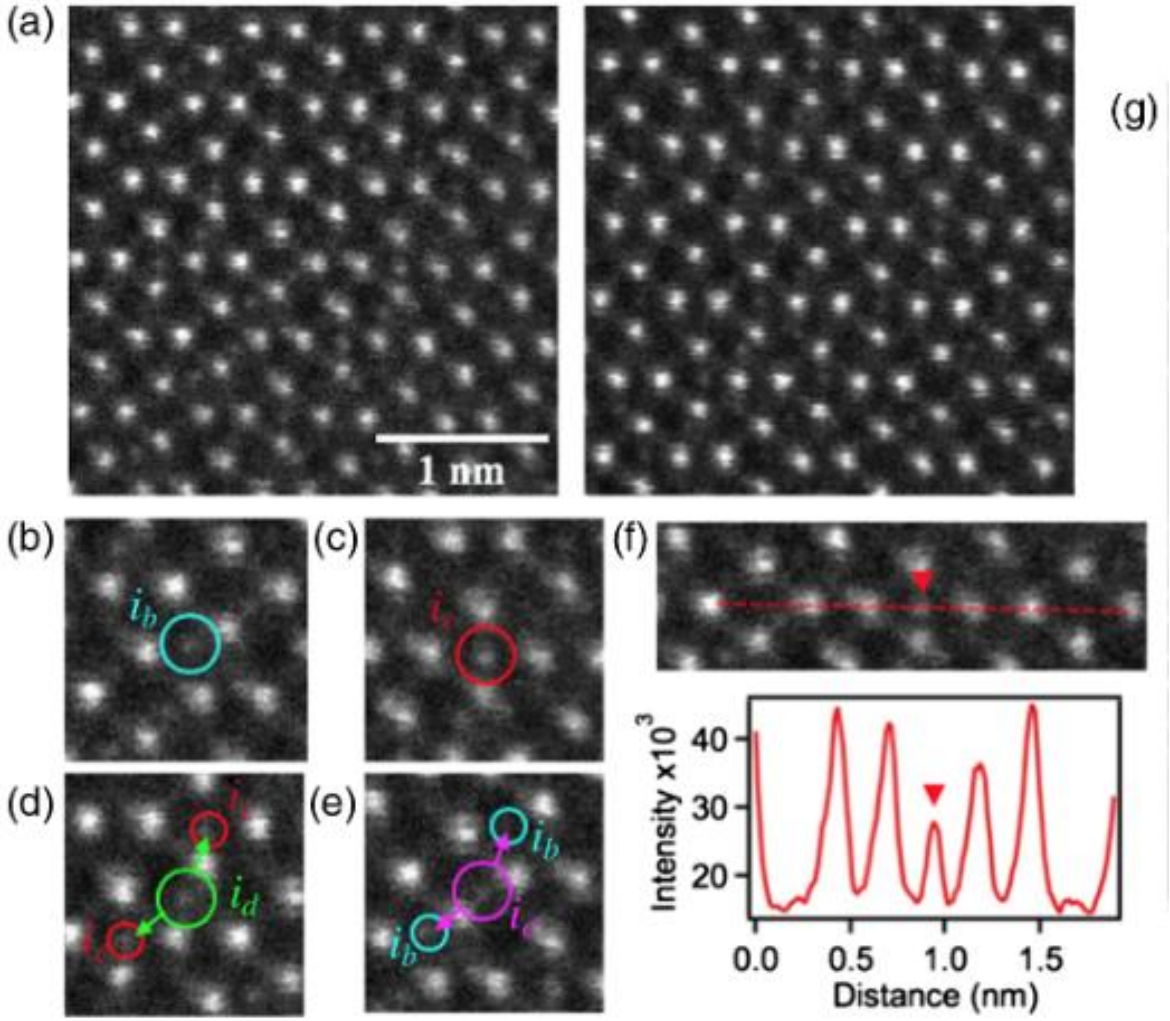


https://link.springer.com/article/10.1007/s10971-022-05779-z

Určovani bodovych poruchy v polovodičových materiálech:

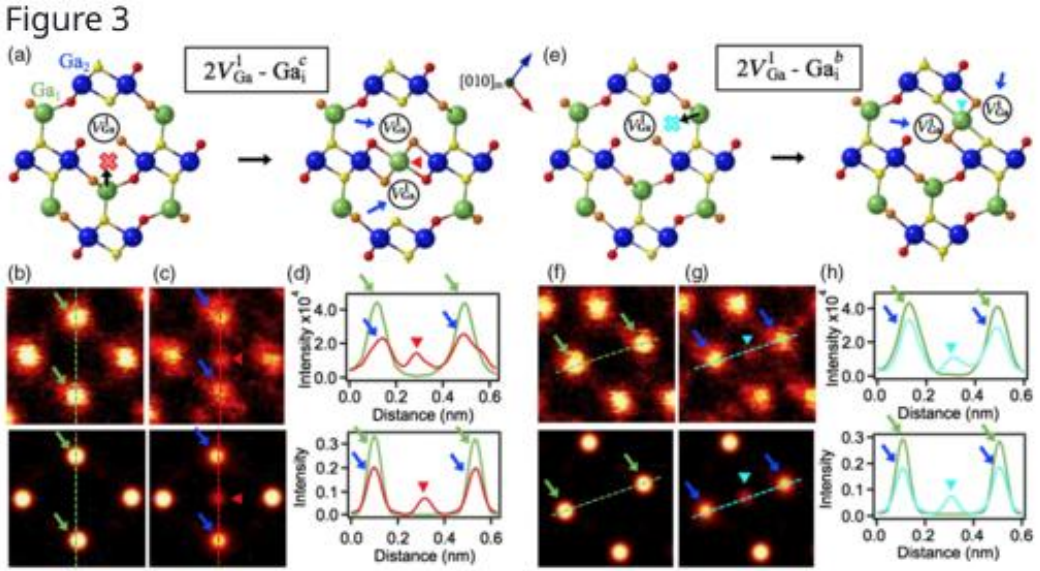
Transmisní elektronová mikroskopie

HAADF-STEM (high angle annular dark field -canning transmission electron microscopy)



β-Ga_2O_3: Sn

i_b, i_c, i_d – *intersticiální atom*



https://doi.org/10.1103/physrevx.9.041027